



June 2, 2025

ASCO Investor Event

Merck & Co., Inc., Rahway, N.J., USA

Forward-looking statement of Merck & Co., Inc., Rahway, N.J., USA

This presentation of Merck & Co., Inc., Rahway, N.J., USA (the “company”) includes “forward-looking statements” within the meaning of the safe harbor provisions of the U.S. Private Securities Litigation Reform Act of 1995. These statements are based upon the current beliefs and expectations of the company’s management and are subject to significant risks and uncertainties. There can be no guarantees with respect to pipeline candidates that the candidates will receive the necessary regulatory approvals or that they will prove to be commercially successful. If underlying assumptions prove inaccurate or risks or uncertainties materialize, actual results may differ materially from those set forth in the forward-looking statements.

Risks and uncertainties include but are not limited to, general industry conditions and competition; general economic factors, including interest rate and currency exchange rate fluctuations; the impact of pharmaceutical industry regulation and health care legislation in the United States and internationally; global trends toward health care cost containment; technological advances, new products and patents attained by competitors; challenges inherent in new product development, including obtaining regulatory approval; the company’s ability to accurately predict future market conditions; manufacturing difficulties or delays; financial instability of international economies and sovereign risk; dependence on the effectiveness of the company’s patents and other protections for innovative products; and the exposure to litigation, including patent litigation, and/or regulatory actions.

The company undertakes no obligation to publicly update any forward-looking statement, whether as a result of new information, future events or otherwise. Additional factors that could cause results to differ materially from those described in the forward-looking statements can be found in the company’s Annual Report on Form 10-K for the year ended December 31, 2024 and the company’s other filings with the Securities and Exchange Commission (SEC) available at the SEC’s Internet site (www.sec.gov).



Opening Remarks

Dr. Dean Y. Li
Executive Vice President and President,
Research Laboratories





Unparalleled
Breadth and depth of activity

35
Phase 3 oncology trials demonstrated
statistically significant overall survival

56
FDA approved indications

>3.4 million
patients treated¹

1. Patients treated with commercially available products
as of December 2024

Proven track record of advancing standards of care in oncology

KEYTRUDA is a foundational immunotherapy with increasing patient impact in earlier stage disease

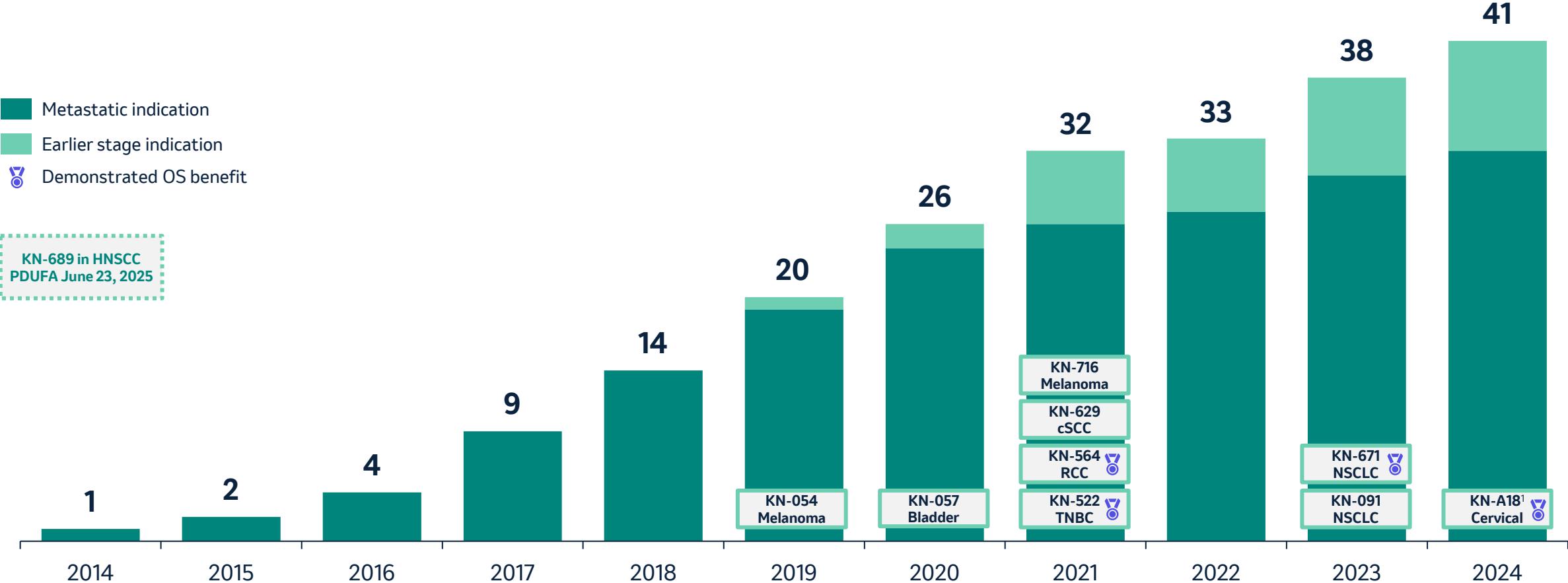


Chart reflects FDA approved indications; 1. OS data not included in label



KEYTRUDA learnings provide roadmap to ongoing and future oncology development



Curative intent setting

Improving access and enabling new paradigms of treatment

PD-1 and chemotherapy combinations

Pursuing next generation chemotherapy (ADCs)

Chemotherapy-free regimens

Replacing invasive chemotherapy regimens with easier to administer alternatives

Less responsive tumors

Leveraging KEYTRUDA learnings to address tumors where PD(L)-1 inhibition has been less effective

Agenda

ASCO and Recent Data Updates | [Dr. Marjorie Green](#)

Development Strategy | [Dr. Eliav Barr](#)

Commercial Opportunity | [Chirfi Guindo](#)

Closing Remarks | [Dr. Dean Li](#)

Q&A | [All](#)



ASCO and Recent Data Updates

Dr. Marjorie Green
Senior Vice President, Head of Oncology
Clinical Development



Subcutaneous pembrolizumab with berahyaluronidase alfa noninferior to IV, with ~2 minute administration in pivotal MK-3475A-D77 trial

MK-3475A-D77

mNSCLC

	Pembrolizumab SC + chemotherapy	Pembrolizumab IV + chemotherapy
Cycle 1 AUC _{0-6wks} (µg·day/mL)	N=245	N=126
Geometric mean (95% CI)	1633.24 (1555.23-1715.15)	1437.58 (1373.68-1504.46)
Geometric %CV	40.4	26.2
Geometric mean ratio (96% CI)	1.14 (1.06-1.22); p<0.0001	
Steady-state (Cycle 3) C _{trough} (µg/mL)	N=202	N=101
Geometric mean (95% CI)	39.23 (37.04-41.55)	23.49 (21.61-25.54)
Geometric %CV	43.3	44.2
Geometric mean ratio (94% CI)	1.67 (1.52-1.84); p<0.0001	

Prespecified noninferiority margin for AUC_{0-6 wks} and C_{trough} geometric mean ratios: 0.8

Data presented at ELCC 2025

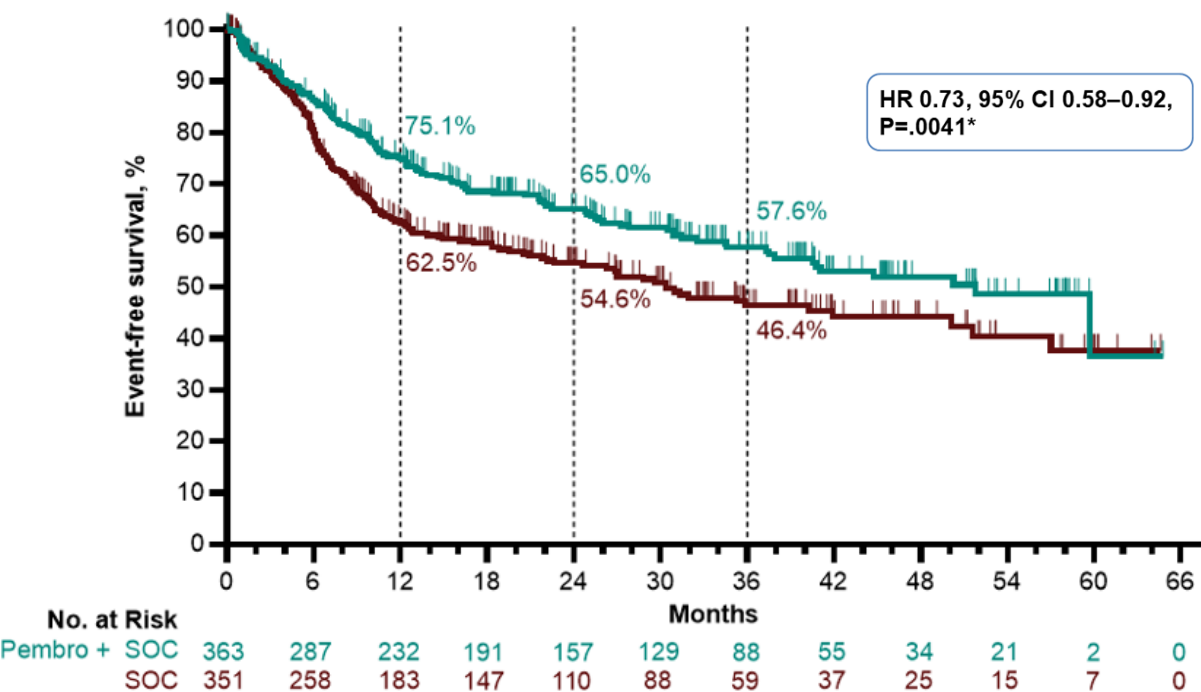
- Subcutaneous pembrolizumab with berahyaluronidase alfa plus chemotherapy demonstrated **noninferior pharmacokinetics** compared to IV pembrolizumab plus chemotherapy
- Secondary endpoints ORR, PFS, DOR and safety **generally consistent between treatment arms**, with median OS not reached in either arm
- Median injection time of ~2 minutes** with Q6W administration, saving time for patients and providers compared to 30-minute IV KEYTRUDA



KEYNOTE-689 further supports ambition of treating patients in earlier stages of disease

KEYNOTE-689

EFS: All participants



*Significance boundary was met at IA1; Data cutoff date: July 25, 2024
Data presented at AACR 2025

- Neoadjuvant/adjuvant pembrolizumab added to SOC significantly **improved EFS** in resectable LA HNSCC in **all prespecified analysis populations**

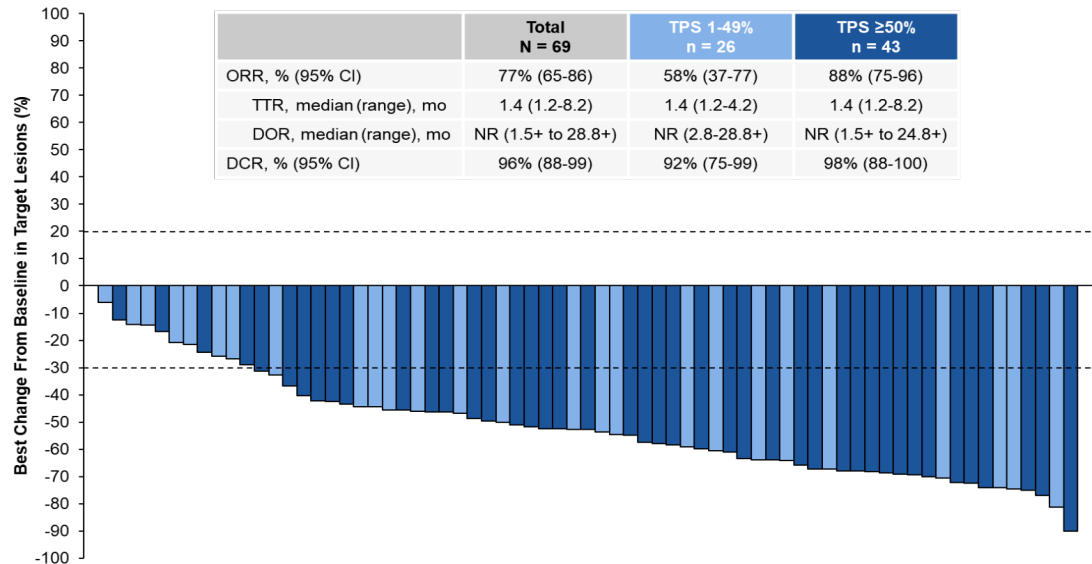
Patient population	Hazard Ratio	CI	P-value
All participants	0.73	0.58 - 0.92	.0041
CPS >1	0.70	0.55 - 0.89	.0014
CPS >10	0.66	0.49 - 0.88	.0022

- **Trend toward improvement in OS** observed in **patients with CPS ≥10** at time of interim analysis



KANDLELIT-001: Encouraging activity for KRAS G12C inhibitor in mNSCLC and mCRC supports best-in-class combination potential

1L mNSCLC - Arm 2: MK-1084 + pembrolizumab

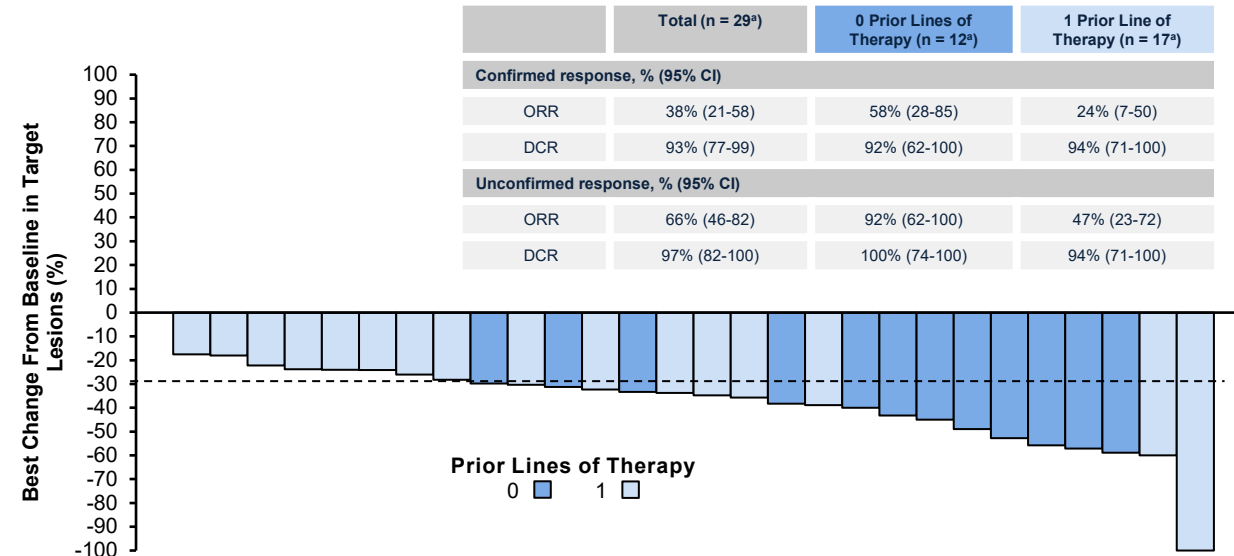


All responses are confirmed. The efficacy population includes all participants with measurable disease per RECIST v1.1 at baseline who received ≥1 MK-1084 dose ≥5 weeks before the data cutoff date. Waterfall plots include all participants who had ≥1 evaluable post-baseline imaging assessment.

- Confirmed **ORR of 77%** with MK-1084 plus KEYTRUDA
- Grade 3-4 treatment-related elevations occurred in **<10% of participants** and were manageable with dose modification and standard medical practice
- Phase 3 studies** of MK-1084 plus KEYTRUDA in KRAS G12C mNSCLC are enrolling (**KANDLELIT-004: PD-L1 TPS >50%**) or planned (**KANDLELIT-007: 1L all comers**)

MK-1084 is in collaboration with Taiho and Astex

mCRC - Arm 6: MK-1084 + cetuximab + mFOLFOX6



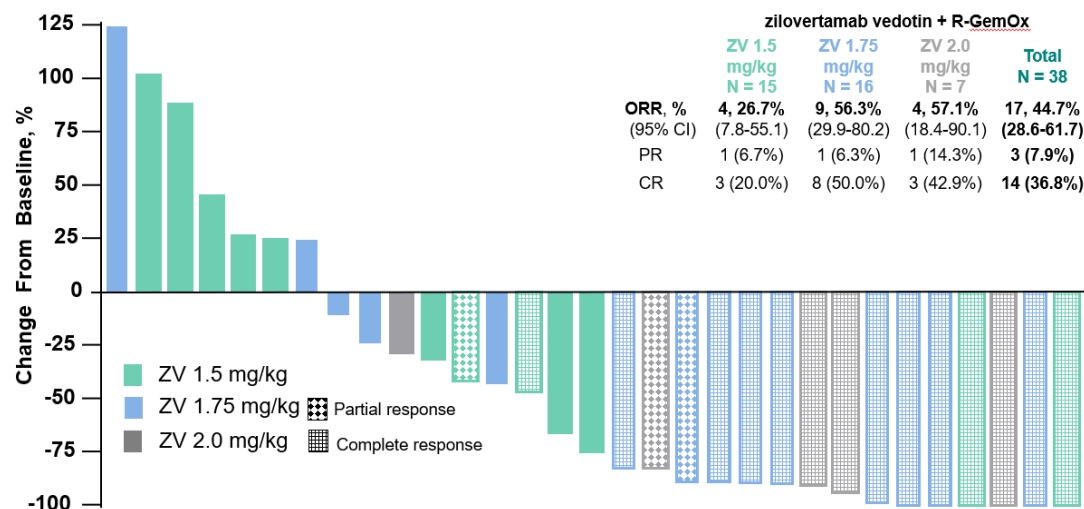
^aResponse was evaluated in the efficacy population, which includes all participants with measurable disease per RECIST v1.1 at baseline who received ≥1 MK-1084 dose ≥5 weeks before the data cutoff date. The waterfall plot include all participants who had ≥1 evaluable post-baseline imaging assessment (n = 28). Data cutoff date: March 12, 2025 (median follow-up, 4.6 months).

- Encouraging ORR as 1L therapy despite limited follow-up: **confirmed ORR of 58%, unconfirmed ORR of 92%**
- Manageable safety profile** and **promising anti-tumor activity**
- Initiating **Phase 3 study** in mCRC (**KANDLELIT-012**)

Zilovetamab vedotin demonstrates encouraging efficacy and safety data in DLBCL

waveLINE-003 (Phase 2 portion)

Best Percentage Change From Baseline SPD for Target Lesions



Safety

7 dose limiting toxicities occurred in all participants

Adverse events, n (%)	zilovetamab vedotin + R-GemOx			
	ZV 1.5 mg/kg N = 17	ZV 1.75 mg/kg N = 16	ZV 2.0 mg/kg N = 7	Total N = 40
All cause	16 (94)	16 (100)	7 (100)	39 (98)
Drug-related adverse events	16 (94)	16 (100)	7 (100)	39 (98)
Serious	4 (24)	5 (31)	4 (57)	13 (33)
Grade 3-4	9 (53)	10 (63)	6 (86)	25 (63)
Drug-discontinued	0	0	2 (29)	2 (5)
Death	0	0	1 (14)	1 (3)

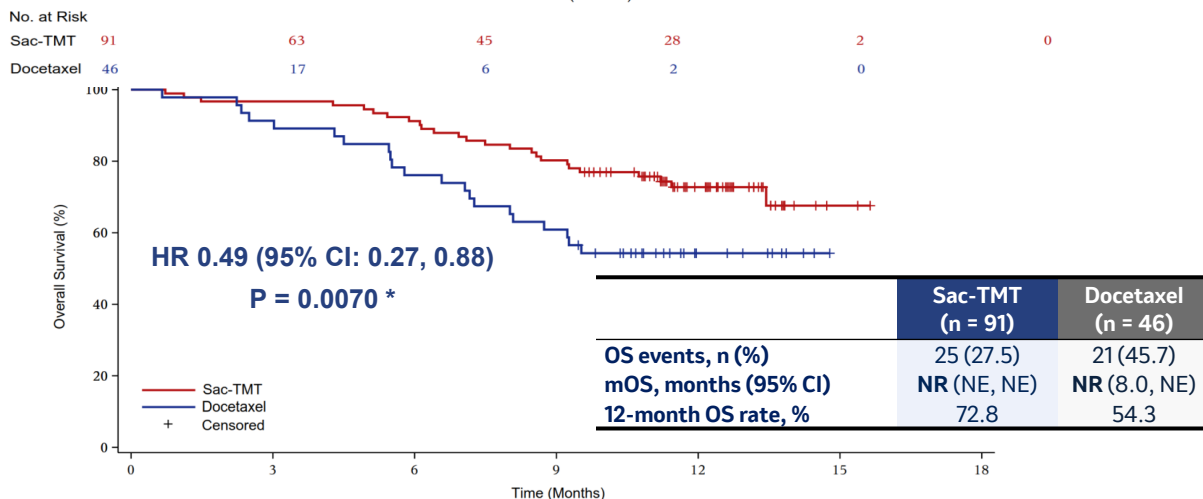
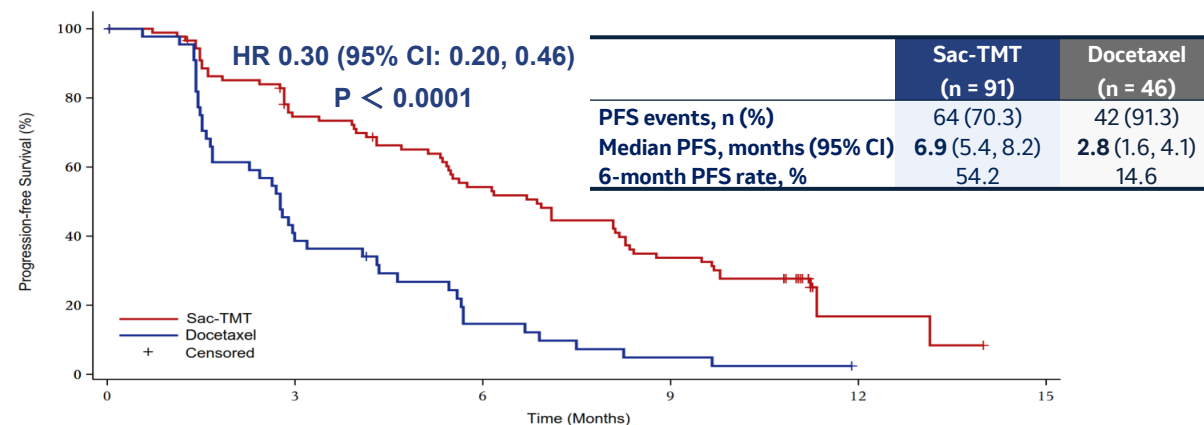
Two participants discontinued ZV due to sepsis or respiratory fatigue (1 in each). One participant died due to sepsis

- **Recommended dose** determined to be **1.75 mg/kg**
- **ORR (56%) and CRR (50%)** of zilovetamab vedotin (ZV) plus rituximab and gemcitabine-oxaliplatin (R-GemOx) at the 1.75 mg/kg dose **improved anti-tumor activity** in r/r DLBCL compared to historical estimates of R-GemOx^{1,2}
- ZV plus R-GemOx demonstrated an **acceptable safety profile**
- ZV also being evaluated for:
 - **1L DLBCL** in a **Phase 3 study** in combination with R-CHP versus R-CHOP (WL-010)
 - **GCB subtype** in a **Phase 2 study** versus polatuzumab vedotin (WL-011)

Compelling data for sacituzumab tirumotecan (sac-TMT) in EGFRm NSCLC support ongoing global Phase 3 studies

Phase 2 OptiTROP-Lung03 Study¹

3L+ EGFRm NSCLC



No. at Risk

Time (Months)	0	3	6	9	12	15	18
Sac-TMT	91	88	83	73	39	2	0
Docetaxel	46	42	35	28	10	0	0

Data cutoff date: Dec 31, 2024. Median follow-up was 12.2 months.

sac-TMT is part of a collaboration with Kelun Biotech. 1. Kelun-led study conducted in China

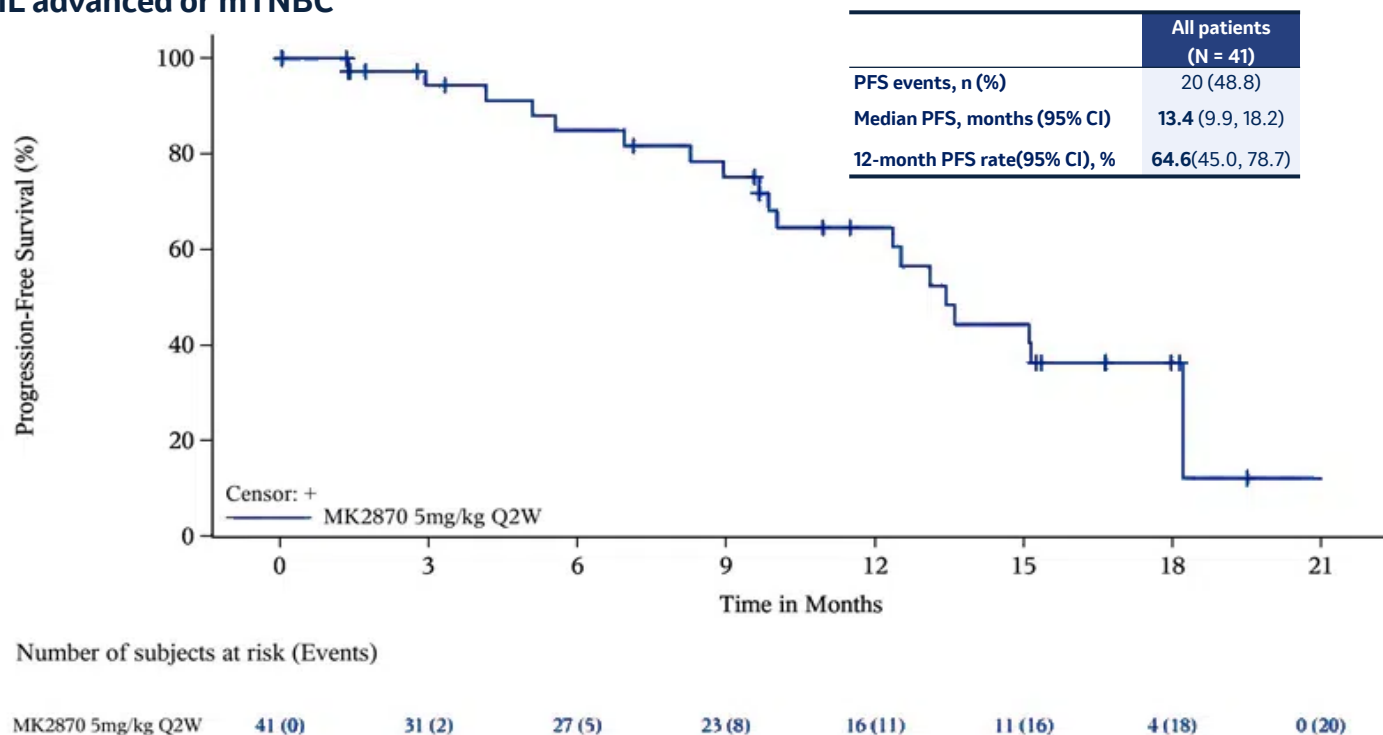
- Demonstrated **statistically significant and clinically meaningful** improvement in **OS and PFS** in previously treated patients with **advanced EGFRm NSCLC**
- Showed **manageable safety profile**, with no unexpected safety signals identified
- **First TROP2 ADC approved for lung cancer** in China based on results of OptiTROP-Lung03 study
- **Initiated global Phase 3 studies** in previously-treated EGFRm NSCLC: TroFuse-004, TroFuse-009



Sac-TMT data in TNBC build on previous evidence and further support ongoing global Phase 3 studies

Phase 2 OptiTROP-Breast05 Study¹

1L advanced or mTNBC



Data cutoff: Nov 18, 2024.

- Demonstrated **promising anti-tumor activity** as 1L treatment for patients with advanced or metastatic TNBC **irrespective of PD-L1 status**
 - **ORR of 70.7%** and **DCR of 92.7%** at 18.6 months of median follow-up
- Showed **manageable safety profile**, with no unexpected safety signals identified
- **Initiated** several **global Phase 3 studies** evaluating sac-TMT in TNBC: TroFuse-011, TroFuse-012, TroFuse-032

Development Strategy

Dr. Eliav Barr

Senior Vice President, Head of Global Clinical
Development & Chief Medical Officer



Uniquely positioned to advance cancer care with differentiated portfolio and pipeline



Immuno-oncology

Boost anti-tumor immune responses

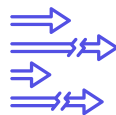
KEYTRUDA[®]
(pembrolizumab) Injection 100 mg

subcutaneous
**pembrolizumab with
berahyaluronidase alfa**
(MK-3475A)

MK-2010
Anti-PD1 x VEGF
bispecific

intismeran autogene¹
(V940)
Individualized
Neoantigen Therapy

**quavonlimab/
pembrolizumab**
(MK-1308A)
anti-CTLA-4



Precision Molecular Targeting

Impact pathways that can drive cancer growth

Lynparza[™]
olaparib
PARACON ON MO

WELIREG[™]
(belzutifan) 40 mg tablets

nemtabrutinib
(MK-1026)
BTK inhibitor

bomedemstat
(MK-3543)
LSD1 inhibitor

LENVIMA[®]
(lenvatinib) capsules | 10 mg and 4 mg
RESULTS THAT MATTER

opevesostat
(MK-5684)
CYP11A1 inhibitor

MK-1084⁴
KRAS G12C inhibitor



Tissue Targeting

Increase cancer cell sensitivity with ADCs and immune-engagers

sac-TMT⁵
(MK-2870)
TROP2 ADC

MK-3120⁵
Nectin-4 ADC

zilovertamab vedotin
(MK-2140)
ROR1 ADC

MK-2750⁵
ADC - undisclosed target

MK-6837
ADC - undisclosed target

MK-6204⁵
ADC - undisclosed target

HER3-DXd⁷
(MK-1022)
HER3 ADC

I-DXd⁷
(MK-2400)
B7H3 ADC

R-DXd⁷
(MK-5909)
CDH6 ADC

gocatamig⁷
(MK-6070)
DLL3 TCE

MK-1045
CD3xCD19 TCE

**Undisclosed preclinical
ADCs^{5,6}**



Our strategy to maximize benefit to patients: unambiguous efficacy, informed combinations, biomarker enrichment, curative intent settings

Best outcomes occur when non-cross resistant therapies are combined or given sequentially



Combinable agents

Develop medicines with single agent activity which can be combined with other agents

sac-TMT, MK-2010, intismeran autogene, MK-1084

Predictive biomarkers enable stratification that can improve therapeutic index of medications



Biomarkers

Improve the therapeutic index through precision medicine

MK-1084, sac-TMT

Early treatment improves chance for better outcomes



Earlier-stage disease

Develop medicines in earlier stages of disease

KEYTRUDA, MK-3475A, intismeran autogene, sac-TMT

Select
Examples



Leveraging unique portfolio to continually advance standards of care for patients

Now

Near

Next

Select examples

Earlier-stage
NSCLC

KEYTRUDA OS benefit in resectable stage II-III

KEYTRUDA + sacTMT¹ or
intismeran autogene² in pCR(-)
patients

Ovarian

KEYNOTE-B96 with PFS and OS
benefit in certain platinum-
resistant patients

KEYTRUDA + raludotatug
deruxtecan (R-DXd)³

NSCLC

KEYNOTE-189 and KEYNOTE-
407 OS benefit in 1L

KEYTRUDA + MK-1084⁴ or
sac-TMT¹

SCLC

Robust independent activity for
ifinatumab deruxtecan (I-DXd)³
and MK-6070³

I-DXd³ and gocatamig³ in
combination

Robust discovery and early phase
pipeline further advances SoC

MK-2010 and other ADCs

1. Collaboration with Kelun Biotech 2. Collaboration with Moderna 3. Collaboration with Daiichi Sankyo 4. Collaboration with Taiho and Astex



Building on foundational regimens with novel combination strategies in earlier stage cancers

More than
30,000 patients
in **>30 active**
Phase 3 trials
in earlier stages
of disease across
10 tumor types

KEYTRUDA-based studies

Lung



KN-091¹

Adjuvant stage IB (T2a ≥4 cm), II or IIIA NSCLC

KN-671¹

Perioperative II, IIIA or IIIB NSCLC

Renal



KN-564¹

Adjuvant intermediate-high or high risk RCC

Skin



KN-054¹

Adjuvant high-risk stage III melanoma

Breast



KN-522¹

Perioperative high-risk earlier-stage TNBC

Bladder



KN-123⁴

Adjuvant locally advanced MIBC

KN-676⁴

Persistent or recurrent HR NMIBC

Ongoing combination studies

intismeran autogene³

INTerpath-002
Adj NSCLC

intismeran autogene³

INTerpath-009
Neoadj/adj NSCLC

sac-TMT²

TroFuse-019
Adj NSCLC

WELIREG

LITESPARK-022
Adj RCC

intismeran autogene³

INTerpath-004
Adj RCC

intismeran autogene³

INTerpath-001
Adj Melanoma

sac-TMT²

TroFuse-012
Post-neoadj TNBC

sac-TMT²

TroFuse-032
Neoadj TNBC or HR-low/HER2- BC

MK-1022⁵

HERTHENA-Breast03
Neoadj TNBC or HR-low/HER2- BC

New since ASCO 2024

1. U.S. indication for certain patients with cancer type 2. sac-TMT, collaboration with Kelun Biotech 3. Collaboration with Moderna 4. Ongoing study, not an approved indication 5. Collaboration with Daiichi Sankyo



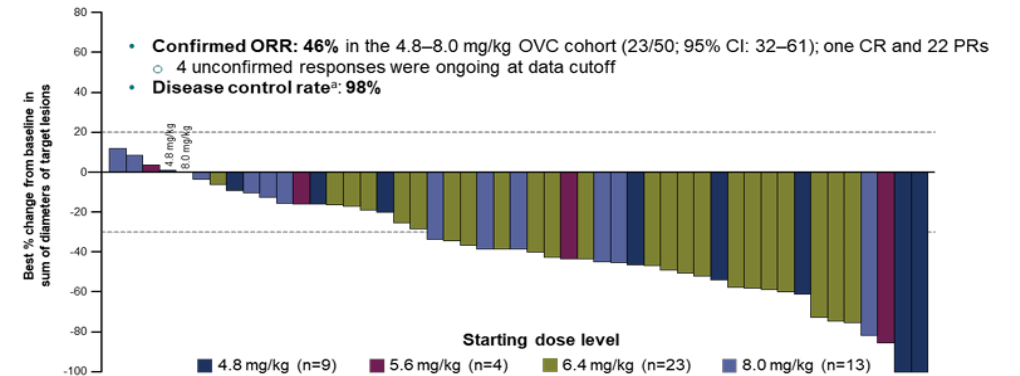
Building on KEYNOTE-B96 with novel combination strategy to improve outcomes for patients with ovarian cancer

KEYNOTE-B96

- KEYTRUDA plus chemotherapy with or without bevacizumab demonstrated **statistically significant and clinically meaningful improvement in PFS** regardless of PD-L1 status and **OS in patients whose tumors express PD-L1¹** for the treatment of patients with platinum-resistant recurrent ovarian cancer
- **First IO-regimen** to show **statistically significant improvement in OS for ovarian cancer**
- **Trial continuing for OS in ITT population**
- Data to be presented at an upcoming medical meeting

MK-5909 (R-DXd)

Phase 1 study: previously treated ovarian cancer



Data cutoff: July 14, 2023
^aCR + PR + stable disease

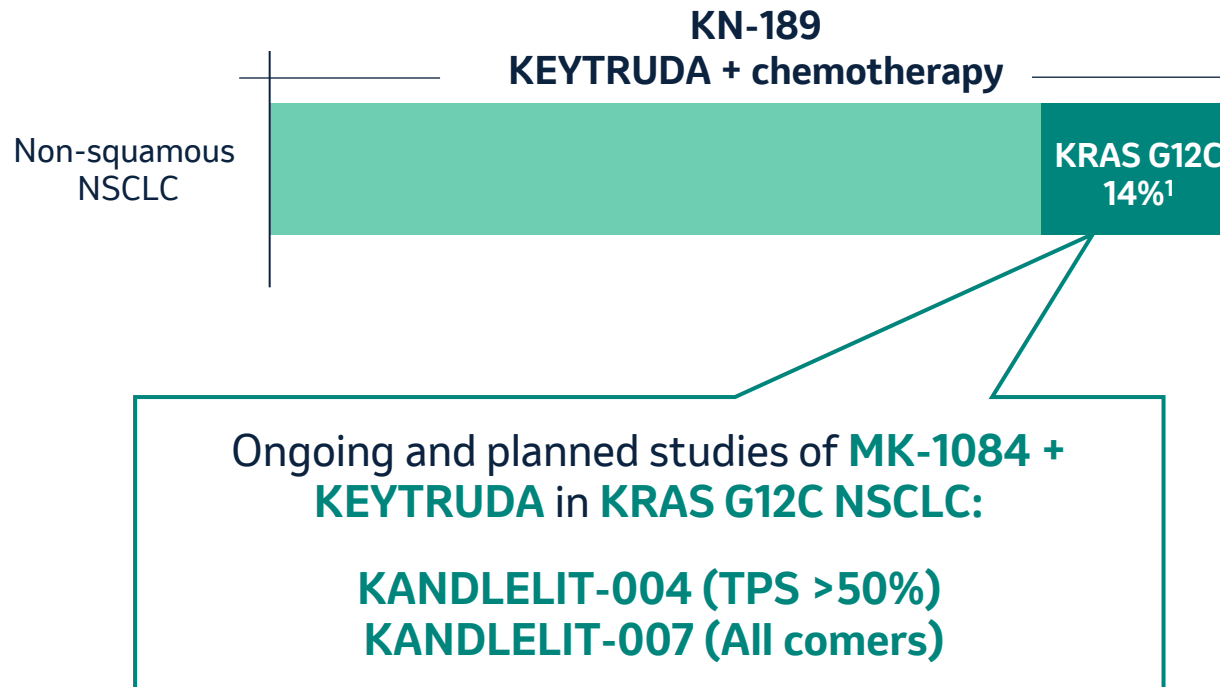
Data presented at ESMO 2023

- R-DXd is **first CDH6-directed ADC** to demonstrate **promising efficacy** in patients with **heavily pre-treated platinum-resistant ovarian cancer** who are **not selected based on CDH6 expression**

Planning underway to evaluate combination of KEYTRUDA and MK-5909 in ovarian cancer

Building on KEYNOTE-189 with novel combination strategy to improve outcomes for patients with NSCLC

MK-1084 + KEYTRUDA



- MK-1084 **selectively targets KRAS G12C containing tumors inhibiting proliferation signaling**
- **Potent, oral** KRAS G12C inhibitor with a **tolerable side effect profile** that offers **combination potential** with KEYTRUDA and other common therapies
- Encouraging Phase 1 data offers potential for **improved ORR and DoR** in KRAS G12C NSCLC
- Opportunity for future **chemotherapy-free, no infusion regimen** in combination with subcutaneous pembrolizumab plus berahyaluronidase alfa

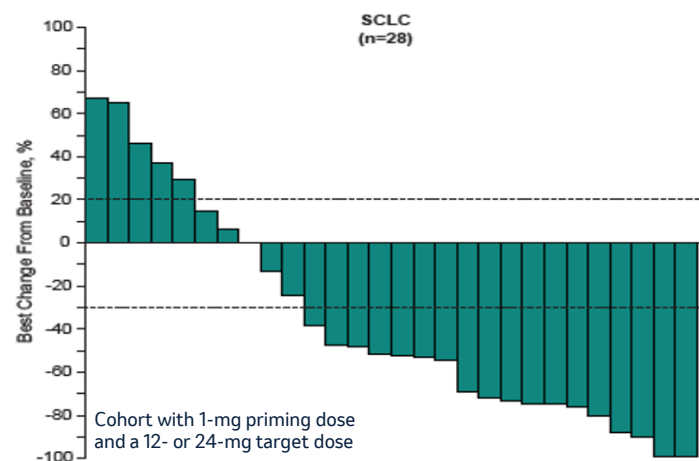
Gocatamig and I-DXd are two highly active agents with potential to improve outcomes for patients with SCLC

- In the U.S., **~34K patients** per year are diagnosed with SCLC¹, representing **~15% of all lung cancers**
- **Median OS** from start of first therapy is **12 months**; **5-year survival** for metastatic SCLC is **~3%**²
- Gocatamig and I-DXd³ are **orthogonal mechanisms** each potentially **highly active** in SCLC
- Both have **manageable safety profiles** with **ability to be combined**
- Combination currently being evaluated in **Phase 1/2 study**

Clinical data

gocatamig

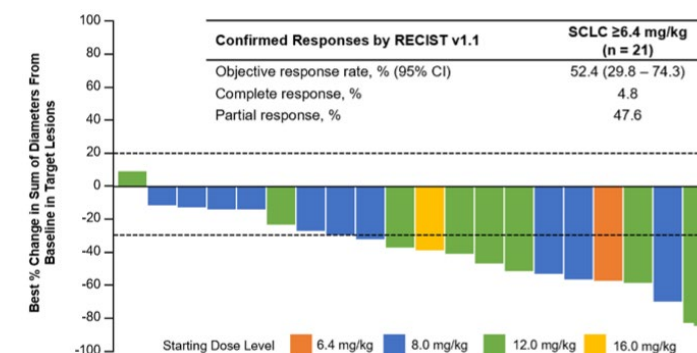
Phase 1 data (SCLC)



Data presented at ASCO 2024

I-DXd³

Phase 1/2 data (SCLC)



Data cutoff: January 31, 2023

Data presented at WCLC 2023

Rapidly advancing one of the industry's broadest ADC programs

	MK-2870	MK-2400	MK-5909	MK-1022	MK-2140	MK-3120	MK-2750	MK-6204	MK-6837
Generic Name	sacituzumab tirumotecan (sac-TMT)	ifinatumab deruxtecan (I-DXd)	raludotatug deruxtecan (R-DXd)	patritumab deruxtecan (HER3-DXd)	zilovertamab vedotin	Undisclosed	Undisclosed	Undisclosed	Undisclosed
Target	TROP2	B7H3	CDH6	HER3	ROR1	Nectin-4	Undisclosed	Undisclosed	Undisclosed
Partner	Kelun Biotech	Daiichi Sankyo	Daiichi Sankyo	Daiichi Sankyo	N/A	Kelun Biotech	Kelun Biotech	Kelun Biotech	N/A
Status	Phase 3	Phase 3	Phase 2/3	Phase 2	Phase 3	Phase 1	Phase 1	Phase 1	Phase 1
Current Tumor Types ¹	Breast, Cervical, Endometrial, Gastric, NSCLC, Ovarian	ESCC, SCLC, Prostate	Ovarian	Breast	DLBCL	Advanced Solid Tumors	Advanced Solid Tumors	Advanced Solid Tumors	Advanced Solid Tumors



Comprehensive development strategy for sac-TMT with potential first-in-class opportunities in majority of study populations



Registrational Study	Tumor Type	Patient Population	Potential First-in-Class	Combo/ Novel Approach	Biomarker Approach
TroFuse-004	NSCLC	3L EGFRm NSCLC			
TroFuse-005	Endometrial	Post platinum & post I/O endometrial cancer	✓		
TroFuse-007	NSCLC	PD-L1 TPS ≥ 50%			
TroFuse-009	NSCLC	2L non-squamous EGFRm NSCLC			
TroFuse-010	Breast	HR+/HER2- unresectable LA or mBC	✓	✓	
TroFuse-011	Breast	LA or mTNBC PD-L1 at CPS <10	✓	✓	
TroFuse-012	Breast	TNBC did not achieve pCR at surgery			
TroFuse-015	Gastric	3L+ advanced/metastatic gastroesophageal adenocarcinoma	✓		
TroFuse-019	NSCLC	Adjuvant NSCLC (no pCR post surgery)	✓	✓	
TroFuse-020	Cervical	2L metastatic cervical cancer	✓		✓
TroFuse-022	Ovarian	Platinum sensitive recurrent OC	✓		✓
TroFuse-023	NSCLC	mNSCLC maintenance treatment (post KEYTRUDA + chemo)	✓	✓	
TroFuse-032	Breast	Neoadjuvant TNBC or HR-low/HER2- breast cancer		✓	
TroFuse-033	Endometrial	pMMR endometrial cancer	✓	✓	✓

Comprehensive companion diagnostic work ongoing, including digital pathology

Series of upcoming pivotal read-outs with >60 registrational trials across 16 assets

Immuno-oncology
Precision Molecular Targeting
Tissue Targeting
✓ PCD passed

KEYTRUDA

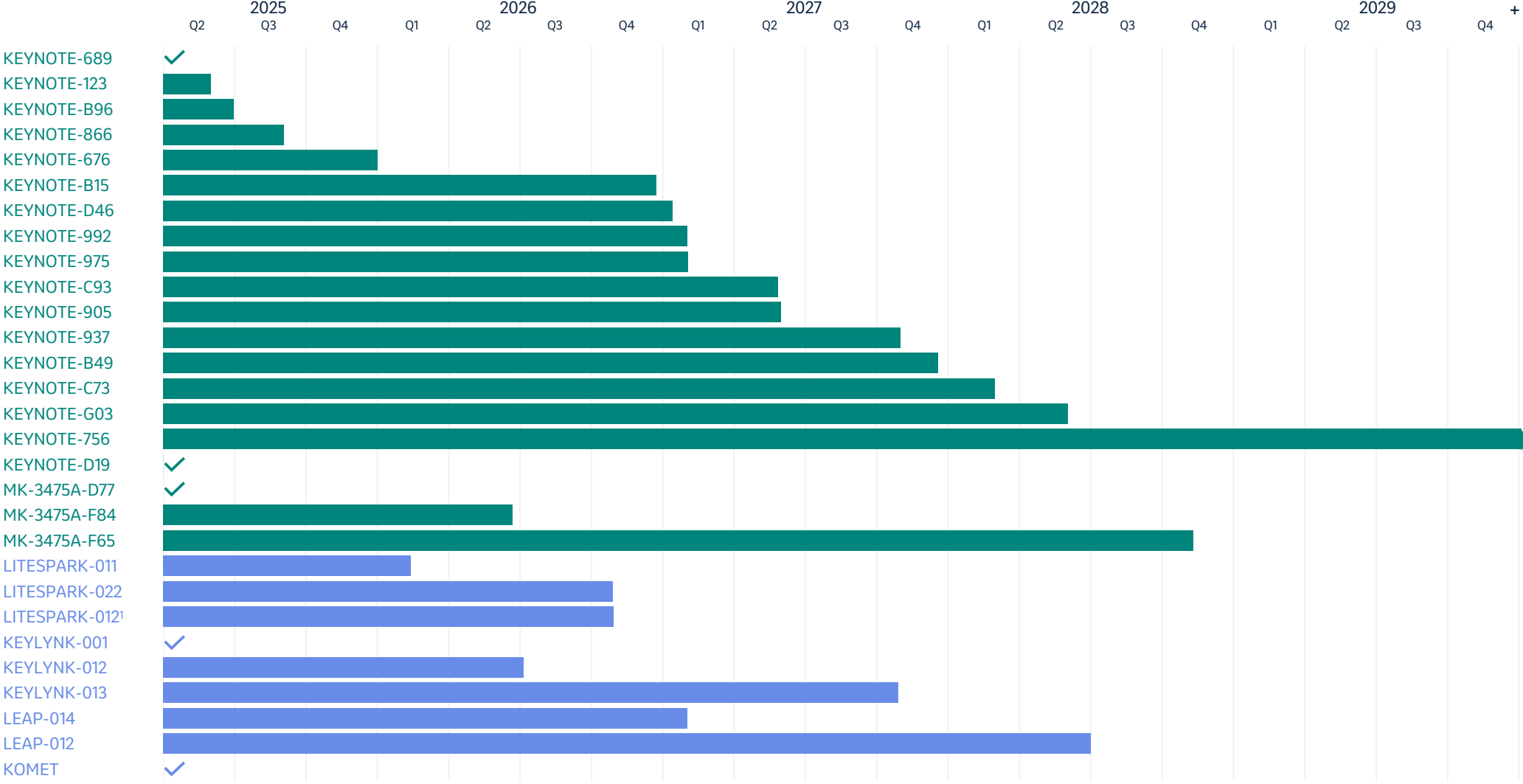
Subcutaneous
pembrolizumab

WELIREG¹

Lynparza

Lenvima

Koselugo



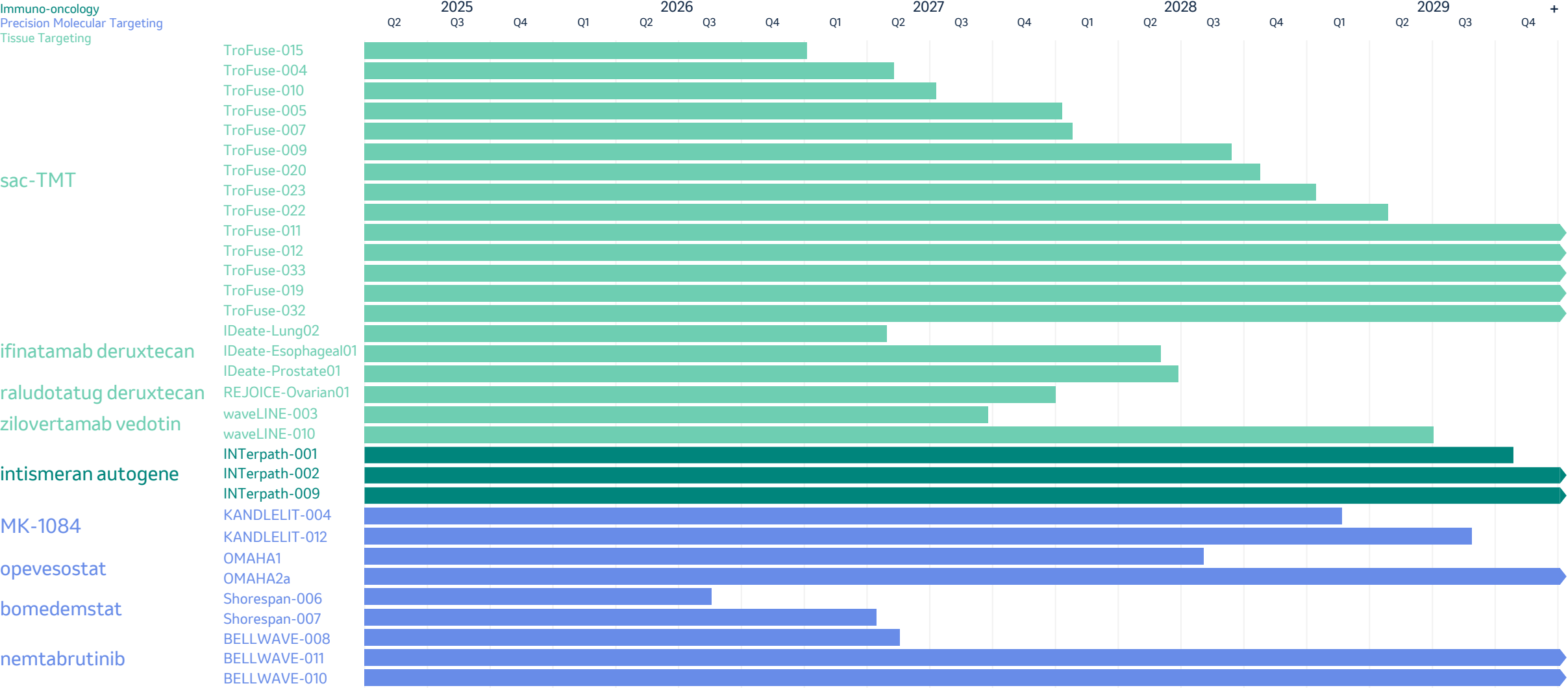
1. LITESPARK-012 is evaluating KEYTRUDA plus WELIREG plus Lenvima or pembrolizumab/quavonlimab (MK-1308A) plus Lenvima

Timing reflects primary completion date, most studies include opportunities for interim analyses



Series of upcoming pivotal read-outs with >60 registrational trials across 16 assets

Immuno-oncology
Precision Molecular Targeting
Tissue Targeting



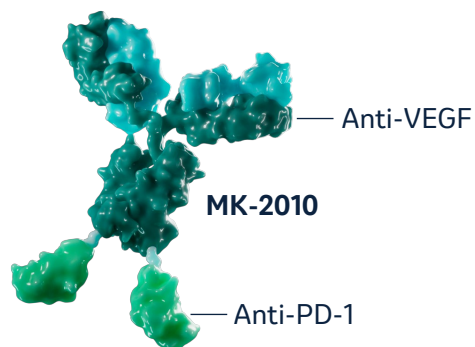
Timing reflects primary completion date, most studies include opportunities for interim analyses



MK-2010: significant opportunity based on promising science of PD1xVEGF bispecific coupled with our unique experience and pipeline

MK-2010 - mechanism of action

- Tetravalent IgG-VHH targeting VEGF and PD-1 with a structure consists of **two IgG VEGF binding sites** and **two nanobodies against PD-1**
- Bispecific format enables **increased PD-1 and VEGF binding**
- No antibody or complement dependent cytotoxicity based on preclinical data
- **Strong inhibition of tumor growth** in humanized mice
- **Well tolerated** in GLP tox



- **‘Advantaged’ company** to fully explore potential of the PD-1xVEGF axis
 - **Leverage PD-1 and VEGF experience** to inform future clinical development program
 - **Portfolio of novel agents** that offer combination opportunities
- Demonstrated ability to advance promising science in oncology with **speed and rigor**
- **MK-2010** currently being evaluated in **Phase 1/2 China-only study¹**
- **Significant opportunity** across a range of tumor types driven by **science** and **unparalleled clinical development capabilities**

1. LaNova-led study conducted in China

Commercial Opportunity

Chirfi Guindo
Chief Marketing Officer, Human Health



Driving patient benefits and global growth with our commercial execution engine



>3.4 million

patients have been treated with commercially available medicines¹



56

FDA approved indications²

44 approvals in EU
41 approvals in Japan

25

tumor types plus MSI-H / dMMR and TMB-H in U.S.

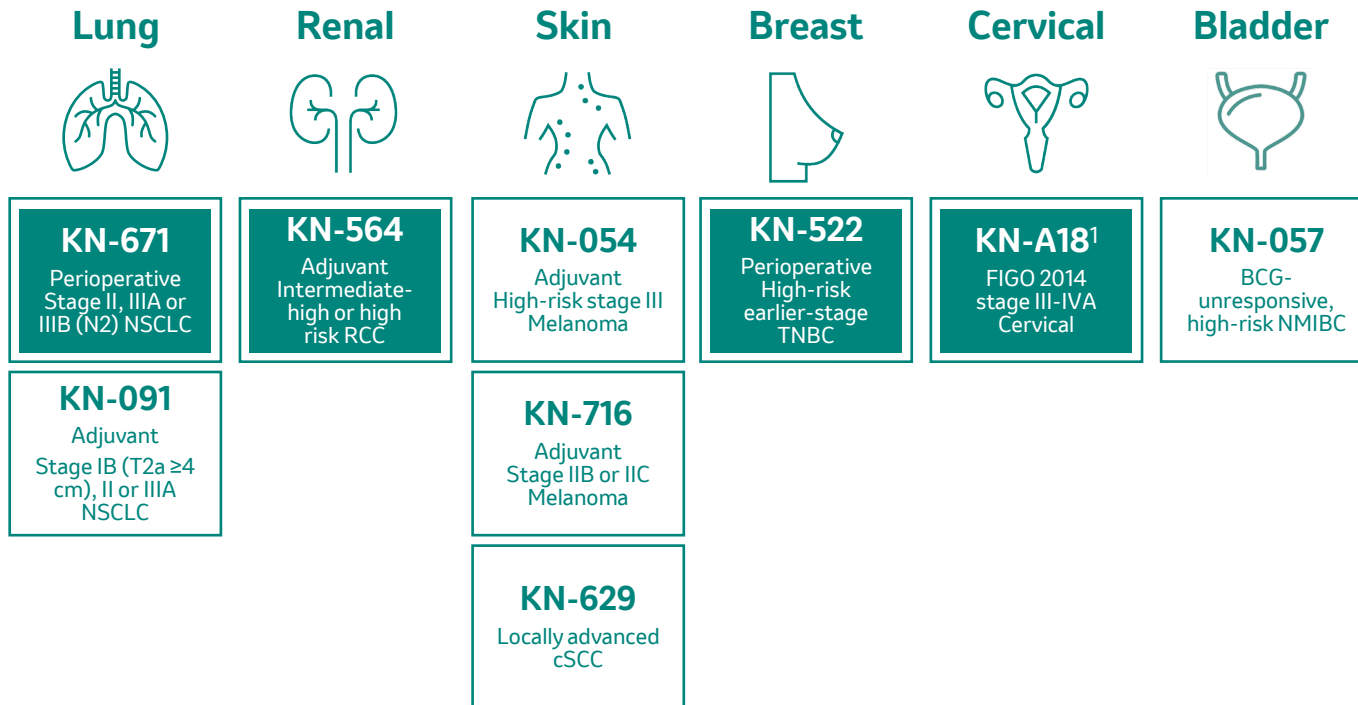
21 tumor types in EU
18 tumor types in Japan



Strong market access capabilities
and reimbursement across HTA markets

In earlier stage disease, KEYTRUDA is the only IO to date to demonstrate significant OS benefit in 4 studies, and to receive 9 approvals

Earlier stage approvals



 U.S. indication for certain patients with cancer type

 Achieved OS in cancer type

1. OS data not included in label

Opportunity in resectable locally advanced HNSCC

HNSCC



KN-689
Perioperative
Resectable LA-
HNSCC

- KN-689 PDUFA June 23, 2025
- First Phase 3 study to demonstrate **statistically significant and clinically meaningful** improvement in EFS in patients with resectable locally advanced HNSCC
- Opportunity for **new standard of care**
- Estimated **>58K new cases of HNSCC diagnosed** and **>12K deaths** in the U.S. in 2024
 - ~**60%** of HNSCC cases are diagnosed as locally advanced

 PDUFA pending



Subcutaneous pembrolizumab may provide important benefits to multiple stakeholders and represents a meaningful global opportunity

Patients



Demonstrated **consistent pharmacokinetic, efficacy and safety profile** to IV KEYTRUDA in MK-3475A-D77 trial

Shortened administration time of ~1-2 minutes vs IV KEYTRUDA with same optionality for **Q3 and Q6 weekly dosing**

Providers and payors



May reduce time spent on treatment-related tasks including administration and prep time¹

Potential to improve access, particularly where infusion center capacity is limited

U.S.



PDUFA **September 23rd 2025**

Seeking approval across **all adult solid tumor indications** for KEYTRUDA

Anticipate **30% to 40% peak adoption in 18-24 months**

- Highest adoption rates in monotherapy or oral combination regimens, including earlier-stage indications
- Adoption in first 6 months impacted by temporary J-code

Global



Expect **approval in major markets starting in 2026**

Anticipate **30% to 40%** peak adoption on average

- Peak penetration will **vary by market**

1. Presented D77 Time and Motion Study ELCC, March 2025

Significant commercial opportunity from late-phase pipeline across tumor types

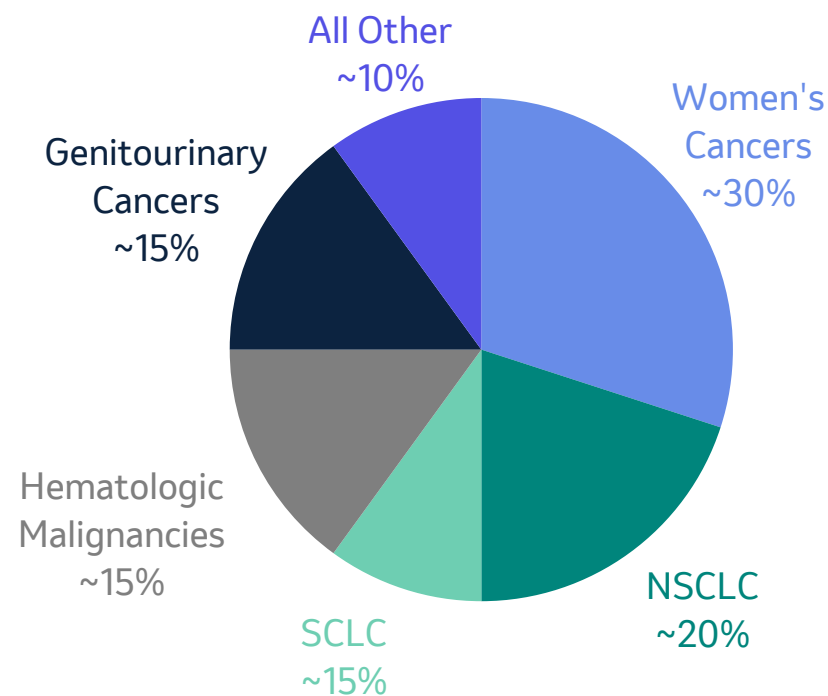
> \$25B

**Pipeline opportunity
by the mid-2030s¹,
more than half in ADCs**

Growing presence in women's cancers
and hematologic malignancies

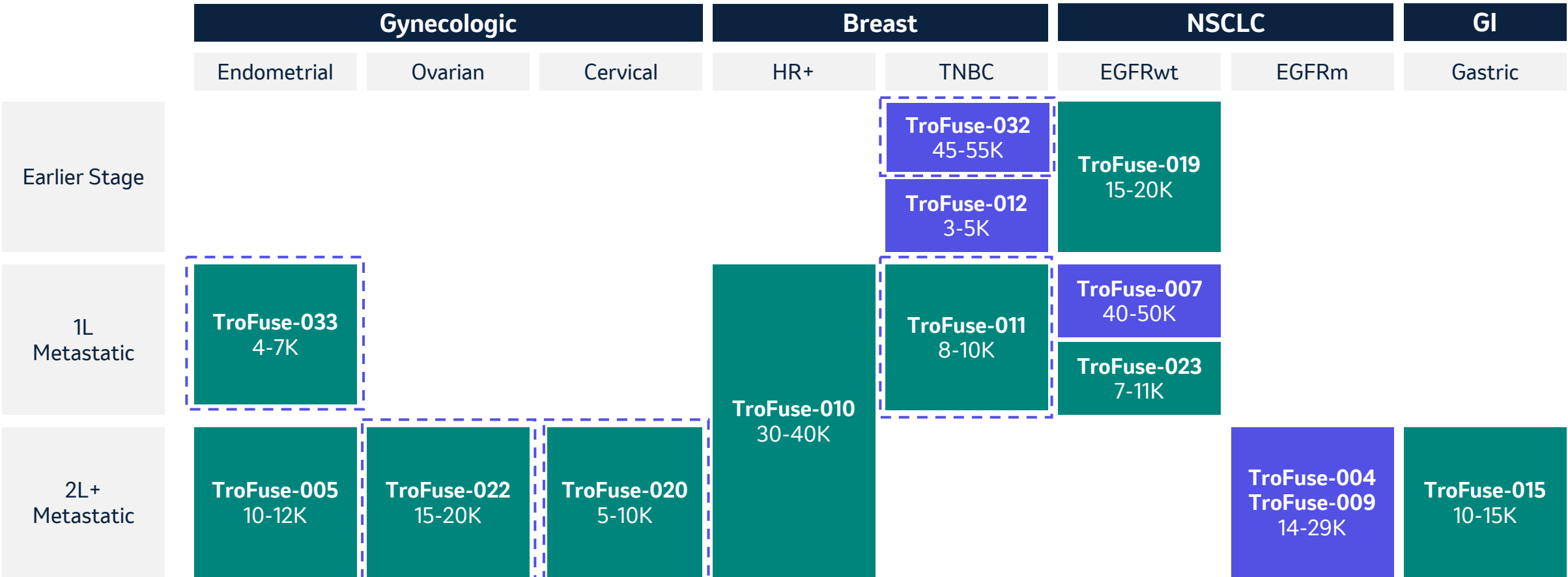
Expanding into new tumor types
(e.g. SCLC, prostate)

Mid-2030s Revenue Opportunity by Tumor



1. Non-risk adjusted annual sales by the mid-2030s; excludes innovation from marketed products, subcutaneous pembrolizumab, early phase programs or additional business development

Sac-TMT addresses a broad range of disease areas and treatment settings with several potential first-in-class opportunities



Study Patient Opportunity¹

First-in-class opportunity

Differentiated approach

New since ASCO 2024

sac-TMT is part of a collaboration with Kelun Biotech 1. Reflects estimated eligible patients based on study population, setting, sequence and regimen; includes U.S., EU5 and Japan

Potential first-in-class portfolio of ADCs with significant revenue opportunity and initial registrational studies underway

	HER3-DXd HER3 ADC ¹	I-DXd B7H3 ADC ¹			R-DXd CDH6 ADC ¹	zilovertamab vedotin ROR1 ADC	
Study	TBA ³	IDEATE-Lung02	IDEATE-Esophageal01	IDEATE-Prostate01	REJOICE-Ovarian01	waveLINE-003	waveLINE-010
Tumor Type	Breast	SCLC	Esophageal	Prostate	Ovarian	DLBCL	DLBCL
Patient Population	TBA	2L ES SCLC	2L+ esophageal	1L+ mCRPC	2L+ platinum resistant ovarian	1L DLBCL	2L+ DLBCL
Patient Opportunity ²	TBA	15 - 20K	25 - 35K	95 - 110K	10 - 20K	55 - 65K	25 - 35K

New since ASCO 2024

1. Collaboration with Daiichi Sankyo; 2. Reflects estimated eligible patients based on study population, setting, sequence and regimen; includes U.S., EU5, Japan and China; 3. Initiation of phase 3 registrational study for HER3-DXd in breast cancer in 2025



Closing Remarks

Dr. Dean Y. Li
Executive Vice President and President,
Research Laboratories



Advancing broad, deep pipeline by leveraging industry-leading capabilities and leadership in IO

Pursuing a focused strategy...

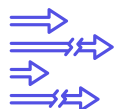
>**60** registrational oncology trials across **16** assets and **3** pillars of biology



Immuno-oncology



Tissue Targeting



Precision Molecular Targeting

+ fully integrated early phase and pre-clinical portfolio

...to profoundly improve patient outcomes

Further improve efficacy

Combine or sequence therapies with different mechanisms

Treat optimally

Utilize predictive biomarkers to identify those most likely to respond

Treat earlier

Develop medicines in earlier stages and curative intent settings





Q&A



Dr. Dean Y. Li
President, Merck
Research Laboratories



Dr. Eliav Barr
SVP, Head of Global
Clinical Development
& Chief Medical
Officer



Dr. Marjorie Green
SVP, Head of Oncology
Clinical Development



Chirfi Guindo
Chief Marketing
Officer, Human Health



Joanne Monahan
SVP, U.S. Oncology



Peter Dannenbaum
SVP, Investor Relations



Appendix

Acronyms

ADC = Antibody-drug conjugate
AE = Adverse event
aRCC = Advanced renal cell carcinoma
B7H3 = B7 homolog 3
BCG = Bacillus Calmette-Guerin
BCMA = B cell maturation antigen
BTC = Biliary tract cancer
BTK = Bruton tyrosine kinase
CD27 = Cluster of differentiation 27
CDH6 = Cadherin-6
CI = Confidence interval
CPS = Combined positive score
CRC = Colorectal cancer
cSCC = Cutaneous squamous cell carcinoma
CTLA-4 = Cytotoxic T-lymphocyte associated protein 4
CYP11A1 = Cytochrome P450 Family 11 Subfamily A Member 1
DFS = Disease free survival
DLL3 = Delta-like ligand 3
DMFS = Distant metastasis-free survival
dMMR = Deficient mismatch repair
EFS = Event free survival
EGFR = Epidermal growth factor receptor
EGFRm = Epidermal growth factor receptor mutant

ES = Extensive-stage
ESCC = Esophageal squamous cell carcinoma
GI = Gastrointestinal
HER2 = Human epidermal growth factor receptor 2
HER3 = Human epidermal growth factor receptor 3
HIF-2 α = Hypoxia-inducible factor-2 α
HR = Hazard ratio
HR+ = Hormone receptor-positive
IA = Interim Analysis
IL-2 = Interleukin-2
INT = Individualized neoantigen therapy
IO = Immuno-oncology
KRAS = Kirsten rat sarcoma viral oncogene homolog
LA = Locally Advanced
LAG-3 = Lymphocyte-activation gene 3
LSD1 = Lysine specific demethylase 1
mCRPC = Metastatic castration-resistant prostate cancer
MHC = Major histocompatibility complex
MIBC = Muscle-invasive bladder cancer
mNSCLC = Metastatic non-small cell lung cancer
MSI-H = Microsatellite instability-high
NMIBC = Non-muscle invasive bladder cancer
NSCLC = Non-small cell lung cancer

ORR = Objective response rate
OS = Overall survival
PARPi = Poly-ADP ribose polymerase inhibitor
PCD = Primary completion date
PD-1 = Programmed cell death protein 1
PD-L1 = Programmed cell death ligand 1
PDUFA = Prescription Drug User Fee Act
PFS = Progress free survival
pNET = Pancreatic neuroendocrine tumors
PPGL = Paragangliomas
RCC = Renal cell carcinoma
RFS = Recurrence free survival
ROR1 = Receptor tyrosine kinase-like orphan receptor 1
SCLC = Small cell lung cancer
sqNSCLC = Squamous-cell non-small-cell lung cancer
TBA = To be announced
TCE = T cell engager
TIGIT = T cell immunoreceptor with Ig and ITIM domains
TKI = Tyrosine kinase inhibitor
TMB-H = Tumor mutational burden-high
TNBC = Triple negative breast cancer
TPS = Tumor proportion score
TROP2 = Trophoblast cell surface antigen-2
VHL = Von Hippel-Lindau



Dean Li, MD, Ph.D.

President, Merck Research Laboratories

Dr. Dean Li serves as executive vice president and president of Merck Research Laboratories. He leads the company's worldwide human vaccines and therapeutics research and development organization.

Since joining Merck in 2017, Dean has held leadership roles in the Translational Medicine and Discovery functions and was appointed to President, Merck Research Laboratories in January 2021. Prior to joining Merck, Dean held positions of increasing responsibility in translational medical research at the University of Utah. Most recently he served as the H.A. & Edna Benning Professor of Medicine and Cardiology, chief scientific officer, associate vice president and vice dean at the University of Utah Health System. From 2015 to 2016, he also served as interim CEO of Associated Regional University Pathologists, one of the United States' largest clinical reference laboratories. During his tenure at the University of Utah, he co-founded several biotechnology companies based upon research conducted in his laboratory, including Recursion Pharmaceuticals, Hydra Biosciences and Navigen Pharmaceuticals.

Dean received his Bachelor's degree in Chemistry from the University of Chicago and his graduate and clinical training at Washington University School of Medicine in St. Louis. Dean is a board-certified cardiologist, a member of the American Society for Clinical Investigation and the Association of American Physicians.



Eliav Barr, MD

Senior Vice President, Head of Global Clinical Development, Chief Medical Officer

Dr. Eliav Barr is senior vice president and head of Global Clinical Development and Chief Medical Officer at Merck Research Laboratories. He leads all late-stage clinical development for Merck's human health portfolio and pipeline.

Prior to his current role, Eliav led MRL's Global Medical Affairs organization expanding Merck's scientific engagement and implementation efforts in oncology, vaccines and infectious diseases. Since joining Merck in 1995, Eliav has held positions of increasing responsibility including leadership roles in oncology and infectious diseases clinical development. He was also previously Therapeutic Area Head for Infectious Diseases and managed product development teams in Oncology and Infectious Disease.

Eliav is a cardiologist by training. He received his undergraduate degree from Penn State University and his medical degree from Thomas Jefferson University. He completed his Internal Medicine residency and Cardiology Fellowship at Johns Hopkins University, and subsequently pursued post-doctoral training at the University of Michigan. Prior to joining Merck, he held a faculty position at the University of Chicago.



Marjorie Green, MD

Senior Vice President, Head of Oncology Clinical Development

Dr. Marjorie Green is senior vice president and head of oncology global clinical development at Merck Research Laboratories.

Marjorie joined Merck from Seagen, where she was senior vice president and head of late-stage oncology, leading clinical development of a diverse portfolio of oncology candidates including multiple antibody drug conjugates. She previously held positions of increasing responsibility at Genentech culminating in her tenure as vice president, Global Product Development, head of breast and gynecologic tumor franchise. Previously, she was assistant professor and medical director of the Nellie B. Connally Breast Center and vice-chair of the Institutional Review Board at the MD Anderson Cancer Center, Houston, Texas. During her tenure at MD Anderson, Marjorie established herself as a nationally recognized clinical expert in the management of breast cancer and the treatment and prevention of associated bone metastases and has authored multiple manuscripts and book chapters on preoperative chemotherapy.

Marjorie received her Bachelor of Arts from the University of Notre Dame and her medical degree from the University of Texas Medical Branch. She conducted an internal medicine residency at University of Virginia School of Medicine and completed fellowships in medical oncology and hematology at the MD Anderson Cancer Center.



Chirfi Guindo

Chief Marketing Officer, Human Health

Chirfi Guindo is chief marketing officer for Merck. He is responsible for leading the development and implementation of the company's long-term strategy for the human health portfolio spanning oncology, vaccines, pharmaceutical and pipeline products.

Prior to this role, Chirfi was executive vice president and head of global product strategy and commercialization at Biogen.

Before joining Biogen in 2017, Chirfi spent more than 25 years with Merck in positions of increasing responsibility in finance, sales, commercial and marketing. During his time with Merck, he led global marketing for Merck's HIV portfolio and also led the company's human health businesses in Canada, the Netherlands and South Africa. Chirfi has been recognized for developing strong talent and forging innovative public-private partnerships that expand access to Merck medicines, while elevating the profile of Merck as a patient-focused company.

Chirfi is a graduate of Ecole Centrale de Paris (France) with a degree in engineering and has a master's of Business Administration from New York University's Stern School of Business.



Joanne Monahan

Senior Vice President, U.S. Oncology

Joanne Monahan is Senior Vice President, U.S. Oncology, Merck. She leads the operations, commercialization and P&L for Merck's U.S. Oncology business, including KEYTRUDA, and oversees partnerships with AstraZeneca and Eisai to jointly market Lynparza and Lenvima.

Beyond oncology, Joanne has had many varied roles during her 30-year career at Merck, including, most recently, spending three years leading the U.S. vaccine business for Merck. Joanne also spent 10 years in primary care, including as Vice President and Global Commercial Lead for the company's cardiovascular and diabetes/NASH portfolios.

In her various roles, Joanne has tackled complex business issues, pioneered the launch of key brands and important indications, led through major market events, and has driven the globalization efforts of important vaccines. For her efforts, Joanne has been recognized with a variety of leadership and marketing awards and received recognition from Merck for her contribution to the business, to customers, and to the growth and development of her people and teams.

Joanne holds a Bachelor of Science in English and Speech Communication from the University of Miami, where she graduated with honors.

